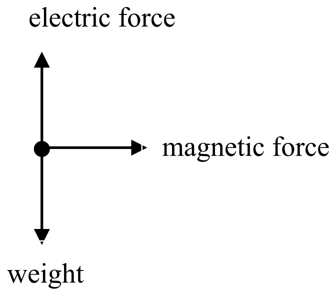
5(a)(i)	The **magnetic force** acting on the charged particle provides the **centripetal force** for the charged particle to move in uniform circular motion.	**B1**
	 $Bqv\sin\theta = ma_c = mv\left(\frac{2\pi}{T}\right)$ $T = \frac{2\pi m}{Bq}$ The circular motion is horizontal, so the net vertical force is zero. $mg = qE$	**C1**
	Combining both equations, $T = \frac{2\pi m}{Bq}$	**C1**
	$= \frac{2\pi E}{Bg}$ (Shown)	**A0**

CKW MC	• Some students stated that both the magnetic force **and** electric force (or sometimes even just the electric force) provide the centripetal force, which was a misconception! Though not credited, an initial free body diagram to identify the correct forces acting on the charged particle in the problem will be a useful starting approach. • The equation based on vertical equilibrium i.e. (1) could not be derived or was not shown separately by quite a number of students. • Some students wrote $F = ma = mg$ for the vertical direction, which is conceptually incorrect application of Newton's 2nd law, since there is no acceleration in the vertical direction i.e. $a = 0$. Thus, $mg = qE$ means that the magnitude of the weight of the particle, mg is equal to the magnitude of the electric force, qE on the particle	
5(a)(ii)	From (i), $T = \dfrac{2\pi E}{Bg} = \dfrac{2\pi(150)}{(0.50)(9.81)}$ = **192 s**	**A1**
CKW MC	This part is well done, except for a few students who make careless mistakes.	
5(b)	When the electric field is removed, the only **vertical force** acting on the charged particle is its **weight**. Hence, the particle ***falls with uniform acceleration*** *g*. The magnetic force still provides the centripetal force for the charged particle to movie in uniform horizontal circular motion, resulting in the charged particle moving in a **helical** path in which the ***distance between adjacent loops increases*** as the particle falls. Using Fleming's Left-hand Rule, the charged particle will continue to move in a **clockwise circle when viewed from the top** of its helical path.	**B1** **B1** **B1**
CKW MC	• Many students use inappropriate terms such as "spiral" or did not state the consequences of a net force due to weight based on Newton's 2nd Law. • Most students did not state the direction of the charged particle in its helical motion and from the correct perspective.	

6 (a) Photons are produced whenever a **charged particle** such as electrons